SUSTAINABILITY



# Closing the Visibility Gap

A joint Kaya AI and Ecolab framework for water stewardship across the data center build-and-operate lifecycle


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# Panel Discussion



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THE VISIBILITY GAP

# The attention to water consumption is in the wrong place - most of a data center's water is spent before the building exists

Average data center water consumption distribution, split by fence line

**Current Focus: Inside the fence** | **Upstream supply chain**

**23%** | **77%**

Reported as Water Usage Effectiveness

- **Cooling tower evaporation**
- **Blowdown and treatment losses**
- **Humidification**
- **Domestic and site use**

**Unmetered, unreported, with no OCP standard.**

**Chips and IT hardware**



**Concrete and steel**



**Chillers, CDUs, gensets**



**Freight and logistics**



**Supplier operations**



*Also outside the fence: water intensity of grid power generation - indirect and ongoing, but part of the footprint*

*Source: CDP Water Security Report, World Resources Institute (WRI) / Corporate Water Stewardship Guidelines*

FRAMEWORKS TODAY

# The approach to tracking Water Usage Effectiveness lags significantly behind both Power and Carbon

Overview of audit / visibility frameworks for Power, Carbon, and Water

| | POWER — PUE | CARBON — CUE | WATER — WUE |
|---|---|---|---|
| **Standard metric exists** | Yes | Yes | Partial |
| **Third-party audited** | Partial | Partial | No |
| **Upstream method exists** | No | Yes | No |
| **Comparable across operators** | Yes | Partial | No |

**Water is also the only one that can stop the building.** In water-stressed regions, data centers can see up to 99% of production and revenue at risk during a disruption. — Ecolab, April 2026

*Source: Ecolab, "Ecolab Unleashes Water Intelligence to Drive Growth in the AI Era," April 2026*

WHERE THE WATER IS

# Only one of four stages in the construction lifecycle that create the water footprint is measured

Overview of water touchpoints across the construction build-and-operate lifecycle

High Visibility  Low Visibility

## 01 SPECIFY

### WATER TOUCHPOINTS

Site water availability, grid power source, cooling topology, and consumptive ratio.

### VISIBILITY TODAY

No water attributes attached to specified products or long-lead equipment.

## 02 PROCURE

### WATER TOUCHPOINTS

Embodied water in chips, IT hardware, concrete, steel, chillers, CDUs, gensets and freight.

### VISIBILITY TODAY

Unmetered and unreported. Substitutions clear on price and lead time, not on water.

## 03 COMMISSION

### WATER TOUCHPOINTS

First fill, flushing, chemical cleaning and passivation before handover.

### VISIBILITY TODAY

Rarely metered separately from construction water.

## 04 OPERATE

### WATER TOUCHPOINTS

Make-up water, blowdown, treatment chemistry, reuse and discharge.

### VISIBILITY TODAY

Metered and reported as WUE. The only stage with a common metric.

**A standard that covers only the operating stage cannot account for the footprint.**

*Source:* BS EN 15978:2011, Sustainability of construction works - Assessment of environmental performance of buildings

OCP GLOBAL SUMMIT | 2026 Scaling Innovation for the AI Era.

HOW THE TWO HALVES FIT

# Supply chain intelligence and water treatment disciplines close the gap together, but help is needed from OCP to set the standard

## SUPPLY CHAIN INTELLIGENCE

KayaAI

- Structures OCP and vendor product data into a queryable knowledge graph: product, specification, revision, conformance claim, supplier, lead time, water attributes.
- Puts those attributes into the buying workflow at design and procurement stage, where the footprint is actually set.
- Carries the same record through to as-built, so reporting reconciles to what was installed.

## WATER SCIENCE INSIGHTS

ECOLAB

- Cooling water chemistry, treatment optimization and real-time system monitoring across operating fleets.
- Predictive leak and anomaly detection on cooling loops, separating true loss from measurement drift.
- Reuse and recycle programs that displace potable make-up water.

## WHERE OCP COMES IN

1. **Define it:** What counts as embodied water, at which boundary, at what data grade.
2. **Convene it:** We can structure and measure. We cannot make suppliers publish.
3. **Make it stick:** A method two companies own is a product. A method OCP owns is a standard.

THE JOINT FRAMEWORK

# Proposed tracking framework: AI-driven supply chain intelligence, joined to water treatment science

Highly preliminary. Illustrative outputs for one specified item, carried end to end.

## 01 MAP

### WHAT IT DOES

Maps water touchpoints across construction and operational supply chains, product by product.

### ILLUSTRATIVE TRACKING

**CDU-07 · 300 kW liquid-to-liquid**

Spec OCP CDU rev 1.2 · OCP Accepted

**Embodied water 1,240 m³ · grade B**

Lead time 42 wks ARO

Kaya AI ●●●●

Ecolab ●●○○

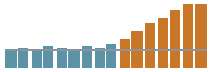
## 02 PREDICT

### WHAT IT DOES

Predictive analytics for leak detection and treatment optimization, built on water chemistry and monitoring.

### ILLUSTRATIVE TRACKING

**CDU-07 · Loop 3 make-up water**



Actual 9% above model, 6 consecutive days

**Loss, not measurement drift — inspect**

Kaya AI ●●○○

Ecolab ●●●●

## 03 REPORT

### WHAT IT DOES

Generates auditable ESG and Net Zero reporting from the same underlying records.

### ILLUSTRATIVE TRACKING

| MODULE | WATER | GRADE |
|---|---|---|
| A1-A3 Product | 1,240 m³ | B |
| A4-A5 Construction | 60 m³ | C |
| B7 Operation | 18,400 m³/yr | A |

*traceable to asset record CDU-07*

Kaya AI ●●●○

Ecolab ●●●○

*Illustrative outputs; all figures are placeholders. Dot rating indicates relative contribution to the capability, not a measu red split.*

WHAT THIS RESTS ON

# Established practice, published evidence

Three published findings: the data gap is real, the obvious fix fails, and carbon has already solved this.

01 THE DATA GAP

## The supplier data does not exist

*“Suppliers would produce full environmental product declarations for all of these components”*

**Julie Sinistore, WSP**
Cradle-to-grave data center LCA, Nature, 2026

**The most rigorous study available says the water data is missing**

02 THE EASY ANSWER

## Waterless cooling only relocates, not erases, the footprint

*“Smart water management begins well before a data center is built”*

**Michael Obradovitch, Ecolab**
Data Center Frontier, November 2025

**Gas generation uses 570-1,100 litres per MWh. Water moves to the grid.**

03 THE PRECEDENT

## Carbon has already solved this problem

*“Data access, data opacity, data grade”*

**OCP Sustainability Project**
Carbon Modeling Workstream

**Water has the same three problems and no workstream.**

*Source: Microsoft/WSP, Nature (2026); M. Obradovitch, Data Center Frontier (2025); OCP Sustainability Project*

CALL TO ACTION

# Standardized water-reuse accounting for next-generation data center facilities

### WHY THIS ALREADY SITS IN OCP

Objective 3 of the OCP Sustainability Project commits to frameworks for supply-side resource consumption across the value chain. Water is a resource. The scope exists; the method does not.

### WHAT WE ARE PROPOSING

This framework as a candidate for community discussion: a common definition of reuse, an agreed lifecycle boundary, and a disclosure format pairing WUE with source and host-utility context.

### WHERE THIS WOULD LIVE

OCP Sustainability Project, alongside the Carbon Modeling and Carbon Disclosure workstreams.

### WHAT WE ARE ASKING FOR

Operators, suppliers and integrators willing to join a scoping call in Q4. Ecolab and Kaya will co-chair the first session.

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# Open Discussion

How should OCP define and account for water reuse across the build-and-operate lifecycle?



# Appendix

DESIGNING FOR OUTCOMES

# When the thermal stack is designed together, water becomes a system outcome

Illustrative comparison: legacy air-cooled hall = index 100



**LIFECYCLE WATER**

**50–70** indexed outcome

30–50% lower lifecycle water consumption versus air cooling in a published cold-plate LCA.

**ENERGY DEMAND**

**≈85** ~15% lower in the same lifecycle study.

**HOW THE OUTCOME IS ENABLED**

- Capture heat at the chip
- Operate warmer, closed loops
- Optimize chemistry, flow and heat rejection
- Preserve optionality for heat reuse

Directional, scenario-sensitive outcomes. See appendix for boundary and evidence.

APPENDIX — COMPARISON METHOD

# How to read the indexed water outcome

The result is a lifecycle comparison, not a universal site-level WUE claim.

### 1 BOUNDARY

Cradle-to-grave water:
- equipment and materials
- electricity generation
- facility operation
- end of life

### 2 BASELINE

- Air-cooled server hall = 100.
- Cold-plate liquid cooling uses the same compute-service boundary and published study assumptions.

### 3 EVIDENCE

Published LCA:
- water: 30–50% lower
- energy: ~15% lower

Result shown as a range because design and location matter.

**WHAT THE INDEX DOES NOT CLAIM**

It does not claim every liquid-cooled site will consume 30–50% less water. Heat rejection, climate, grid mix, water source, utilization, coolant loop design and reuse strategy can materially change the result.

**Stewardship implication: compare architectures before specification, then reconcile design assumptions to as-built and operating data.**

Sources: Microsoft/WSP lifecycle assessment in Nature (2025); Microsoft Source summary (2025).